

Task 1: Histogram plotter (Rainfall figures)

You are asked to write a program (in functions) that will read in twelve integers (all between 0 and 40) which are monthly figures of some imaginary place. You will have to read these figures into an array of integers.

Example: for the following sample rainfall figures:

12 30 10 5 17 15 22 12 11 16 0 21

The histogram drawn will be:

[illegible]

Which is made up of a series of *'s.

For a start write the program without functions first.

Hint:

To go through each item in the array, the loop is:

```
for (int i=0; i < ARRAYSIZE; i++)
{
    ...
}
```

And to draw a line of n asterisks, the code is:

```
for (int j=0; j < Rain[i]; j++)
{
    cout << 'x';
}
```

After you have finished writing the program without functions, put the part that gets the rainfall into a function and the part that prints the histogram into another. Also write a function to tell the user what the program does.

The function declarations should be:

```
void TellUser();
void GetRainfall(int []);
void DrawHistogram(int []);
int Calc total(int []);
```

Task 2:

Create an array and store the following Student marks 68, 72, 78, 69, 85, 95, 75, 77, 82, 84, 91, 89, 95, 74.

From the following set of marks.

1. Calculate the mean

Display the mean at the end of the program.

Task 3:

Declare an array of integers of Size 10. Accept ten integer numbers into the array and try working out the maximum and minimum value entered into the array.

Steps:

1. Create a for loop to read values from the array
2. Create for loop to search for highest value you will have to use if structure for this.
3. Create for loop to search for lowest value you will have to use if structure for this.